

Hariton Pitsik

ML ENGINEER · AI RESEARCHER · DATA SCIENTIST

Saratov, Russia

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Education

Saratov State University

BACHELOR | FUNDAMENTAL COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Saratov, Russia

September 2023 - June 2027

Skills

Programming languages

Python, Rust, SQL

Spoken languages

Russian (native), English (B1-B2)

Technologies

Git & GitHub, Docker, Linux, LaTeX, Typst, Jupyter, NumPy, Pandas, Matplotlib & Seaborn, Scikit-learn, XGBoost, CatBoost, FastAPI, ClearML, Streamlit, PyTorch, Transformers, PostgreSQL

Concepts

Linear Algebra, Mathematical Analysis, Probability Theory & Statistics, Computer Science, Classical ML Algorithms (Supervised & Unsupervised), Deep Learning, Natural Language Processing (NLP), Large Language Models (LLM)

Some achievements

Teaching & Social

- Participant of Development Students Club (DSC)
- I believe education is important and should be shared. I teach machine learning as part of a DSC community.
 - Lecture recordings on NumPy ([YouTube](#)) and Pandas ([YouTube](#)).
 - Machine Learning Club playlist: ([YouTube](#))

Competitions

- Participant in the HSE & X5 Growth Gradient contest, focused on time series forecasting ([Link](#)).
- Participant in the FSP hackathon as an analyst of existing hashing methods and potential sources of entropy.

Projects

Local RAG System

🔗 haritonn/pdf_rag

- Built a fully local RAG assistant that answers questions over uploaded PDF documents;
- Made retrieval configurable from the UI: switchable LLM, embedding model, vector database, and Top-K sources;
- Made answers auditable by exposing the source documents and retrieved contexts in a Streamlit interface; designed the system around scalable abstractions and Qdrant.

Transformers from Scratch

🔗 haritonn/transformers-scratch

- Implemented the full encoder-decoder architecture in PyTorch;
- Built a modular training pipeline for seq2seq tasks;
- Designed a fully customizable pipeline for experiments.

Caption Generator

🔗 haritonn/caption_gen

- Engineered an end-to-end image-captioning pipeline on Flickr8k: ResNet-50 encoder + soft-attention LSTM decoder in PyTorch;

- Improved training stability with scheduled sampling, label smoothing, attention regularization, gradient clipping, and early stopping;
- Built a reproducible experiment workflow with deterministic train/validation/test splits, checkpointing, and optional ClearML tracking;
- Evaluated caption quality with BLEU and METEOR; reached 0.1655 BLEU and 0.3774 METEOR on the validation split (37m. weights).

cargo-smi — NVIDIA GPU Monitor

 [haritonn/cargo-smi](#)

- Developed a responsive terminal dashboard in Rust for unified monitoring of NVIDIA GPUs and host-system resources;
- Integrated NVML to display per-GPU temperature, utilization, VRAM, CUDA/driver versions, and GPU-consuming processes;
- Added multi-GPU navigation, configurable auto-refresh, manual refresh, and a 120-sample utilization history chart;
- Surfaced CPU, RAM, swap, and the 20 most CPU-intensive processes in the same TUI using Ratatui, Crossterm, and Sysinfo.

ResNet with Attention

 [haritonn/resnet_attention](#)

- Implemented baseline ResNet-50 and a channel-attention variant from scratch in PyTorch;
- Improved classification quality over the baseline by adding channel attention; compared convergence, accuracy, and F1 score;
- Built a reproducible training and inference workflow with early stopping, attention visualizations, and ClearML experiment tracking.

Coursework

2nd Year: Generative Computer Vision

 [haritonn/coursework2](#)

COMPARED DIFFUSION MODELS, GANS, AND LARGE MULTIMODAL MODEL (LMM)-BASED APPROACHES FOR VIRTUAL TRY-ON. EVALUATED VISUAL QUALITY AND INFERENCE TIME ON A CUSTOM DATASET.

3rd Year: Text-Embedding Geometry & HDBSCAN Clustering

 [haritonn/coursework3](#)

INVESTIGATED HOW EMBEDDING GEOMETRY AFFECTS NEWS-EVENT CLUSTERING: RAN 2,604 HDBSCAN EXPERIMENTS ACROSS 3 EMBEDDING FAMILIES, 93 FEATURE SPACES, AND 28 PARAMETER VALUES. SHOWED THAT UMAP IMPROVED THE BEST F1 FOR EVERY EMBEDDING FAMILY; ACHIEVED $F1 = 0.837$ WITH JINA EMBEDDINGS V5 + UMAP(10).